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[18MSCH1H02]

FIRST SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER 2019

ORGANIC CHEMISTRY - I

Time :03Hours **Maximum Marks:70**

Instructions: 1. **Question 1 is compulsory, carries 10 marks.**
2. **Questions 2 to 7 carry 15 marks each, answer any four.**

1. Answer any FIVE of the following questions: **5×2=10**

a. Illustrate three forms of Newmann projections with examples.

b. Describe conformational analysis in 1, 4-disubstituted cyclohexan.

c. What is an anti aromaticity?

d. What are carbocations? Explain its structure.

e. What are electrophiles and nucleophiles? Give examples.

f. What is chloromethylation reaction? Give an example.

2. a. i. Explain the Markovnikov's and Anti-Markovnikov'srule with reaction and mechanism.

ii. Give types of sterioisomerism with examples.

iii. Illustrate the different forms of Newmann projection. **(2+5+3)**

b. What are diasteriomers? Mention the types of diasteriomers with examples. **05**

3. a. Describe the conformational analysis in cyclohexane and cyclopentane. **10**

b. What are biphenyls? Write the conditions for optical activity in biphenyls. **05**

4. a. i. What is thermodynamically and kinetically controlled reaction. **05**

ii. Explain the Curtin-Hammett principle. **05**

b. Explain the stability of benzene and cyclobutadiene using molecular orbital theory. **05**

5. a. i. Give any four methods for the formation/generation of carbocations. **05**

ii. What are the factors affecting the stability of carbocation. Explain with an example. **05**

b. Write an account on the formation of carbanions. **05**

6. a. Why do additions to alkenes work? Briefly discuss the addition of HX to alkene. **10**

b. Discuss the energy profile diagram of S_N^1 and S_N^2 reactions. **05**

7. a. What is Chugaev elimination reaction? Explain its mechanism. **10**

b. What is Friedel- Crafts alkylation? **05**

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